

The 5 Biomarkers That Actually Moved the Needle in My CCI + MCAS Recovery

(And the Ones Most Doctors Ignore)

*A no-bullshit guide from someone still in the fight.
Real data. Real terrain. Real proof.*

I'm 30. I have craniocervical instability at C1-C2, suspected hypermobile Ehlers-Danlos, and MCAS-type mast cell activation that was wrecking my life.

For seven months I dealt with flushing that came out of nowhere. Heat sensitivity so bad I couldn't function. Widespread inflammation. And this weird "skin stuck" feeling that made my whole body guard and knot up. My anxiety was through the roof. Brain fog got so bad I had to stop Adderall. I was disappearing from my own life.

Most doctors wanted to throw antihistamines or anxiety meds at single symptoms and call it a day. I wanted the actual upstream drivers fixed.

After getting real imaging (X-rays + TyTron scans) and starting precise upper cervical work, I realized something: the biomarkers that actually tracked my improvement were not the ones most doctors order.

Here are the five that have mattered most to me — and why they're worth tracking if you're dealing with the same neck-vagus-mast cell-guarding loop.

The 5 Biomarkers

1. Morning HRV (Heart Rate Variability) – Your Vagus Brake Gauge

This is the single best daily proxy for how well your vagus nerve is functioning.

When my upper cervical instability was flaring and irritating the vagus, my morning HRV was consistently low and erratic.

As the adjustments started holding and I added long-exhale breathing + mineral support, my HRV rose and stabilized.

How to track it: Free or cheap phone apps (camera on finger). Track first thing in the morning before caffeine or screens.

What improved for me: From chaotic low 30s–40s to consistent mid-50s+ within weeks of consistent terrain work + adjustments.

2. TyTron Paraspinal Thermal Pattern (or any autonomic infrared scan)

This was the most objective proof I had that my nervous system was dysregulated.

The wavy, asymmetrical yellow lines on my scans showed clear upper cervical nerve interference.

After starting Blair adjustments, the patterns began straightening. This is direct visual feedback on whether the mechanical driver is actually being addressed.

How to get it: Upper cervical specialists (like Dr. Chism) often include this. It's cheap compared to MRIs and far more functional for this terrain.

Why it matters: It proves the vagus-neck loop is real and gives you before/after data that standard bloodwork misses.

The 5 Biomarkers (continued)

3. Tryptase + Urine MCAS Mediators (during/after a flare)

Baseline tryptase is often normal in MCAS. The real signal comes from catching it during or right after a flare, plus urine markers (N-methylhistamine, LTE4,

These showed me that my mast cells were over-firing because the vagus brake was off due to neck instability.

How to get them affordably: Ask your doctor for “suspected MCAS workup – baseline tryptase + urine mediators.” Many insurance plans cover it when you tie it to flushing,

What it gave me: Objective confirmation that this wasn't "just anxiety" or "just stress."

4. ESR + CRP (Systemic Inflammation) + Specific Symptoms Tracking

Standard inflammatory markers plus daily tracking of my “guarding score” (how stuck my skin and muscles felt on a 1–10 scale) and flushing/heat episodes.

When the neck stabilized, both the blood markers and my subjective guarding score dropped together.

How to track: Basic bloodwork (usually covered) + a simple daily note in a journal or the health-tracker tool I built.

Why it matters: It connects the mechanical fix to the downstream inflammation you actually feel.

5. Vitamin D + Magnesium Status (Terrain Foundations)

Low Vitamin D and magnesium are extremely common in this presentation and make everything worse — ligament laxity, mast cell reactivity, vagal tone, and

Getting these optimized was foundational before anything else moved.

How to test: Standard bloodwork for 25-OH Vitamin D and RBC magnesium (more accurate than serum).

What changed: Once these were in range, my nervous system became noticeably less reactive and I held adjustments longer.

What to Do Next

Track these five consistently for 30–60 days while you stabilize the mechanical piece (upper cervical work) and the terrain basics (vagal breathing, minerals, gentle movement, sleep, hydration).

You don't need to be perfect. You need data.

I built a simple CLI tool called health-tracker that lets me log symptoms, meds, and daily metrics in seconds. Use whatever system keeps you consistent.

If you want the exact daily tracking template I use, the full PCAC framework, and the raw updates from my own adjustments and labs, join me on Substack at Dose of Proof.

This isn't polished wellness content.

This is proof from someone still in the middle of it.

You're not broken. You're not crazy.

You just haven't had the right map yet.

Join me on Substack — let's build the right map together.

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